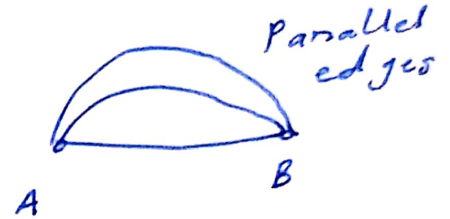
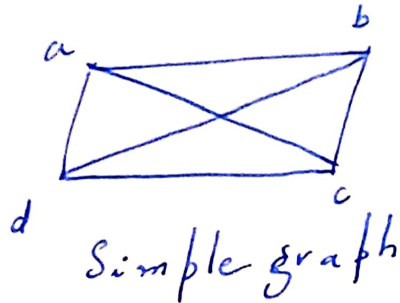
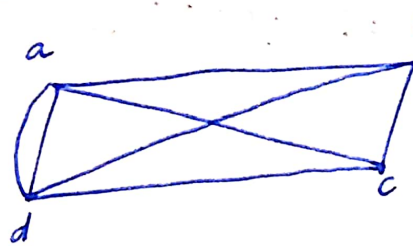


Graph Theory

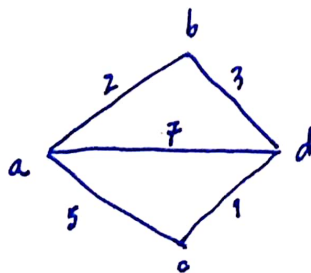
- Simple Graph: A graph with no loops, no parallel edges is called a simple graph.
- Loops: An edge whose starting and ending point is same.



- Multi Graph: A graph with parallel edges is called as multigraph.

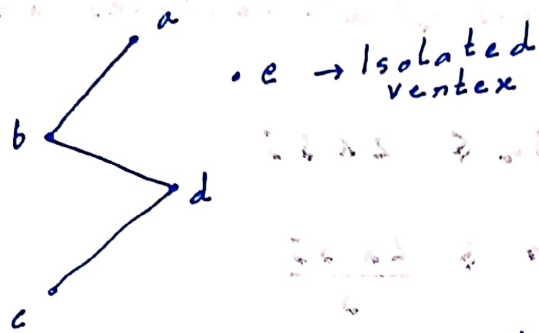


- Null graph: A graph for which edge set is empty is a null graph.
 $E = \phi$
- Trivial graph: A graph with 1 vertex and no edges.
- Weighted graph: A graph in which weights are assigned to every edge is called weighted graph.



□ Degree of a vertex: The number of edges incident on a vertex is called as degree of vertex.

□ Isolated vertex: A vertex of degree zero in a graph is called isolated vertex.

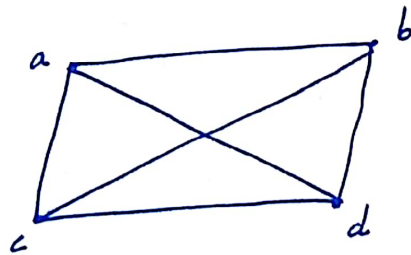


□ Pendant vertex: Vertex with degree 1 in a graph is called as pendant vertex.

□ K-regular graph: A graph is said to be K-regular if every vertex of G has degree 'K'.



□ Complete graph: A simple graph ' G ' in which every pair of distinct vertices are adjacent (i.e. every vertex is connected with every other vertex).



Complete graph with n -vertices $\rightarrow (n-1)$ regular graph.

□ Order of a graph: No. of vertices in a graph.

□ Size of a graph: No. of edges in a graph.

□ Sum of degrees of vertices in an undirected graph = $2 \times (\text{no. of edges})$

The number of vertices of odd degrees in a graph is always even.

Q. What is the largest no. of vertices in a graph with 35 edges if all vertices are of degree atleast 3.

$$3n \leq 2 \times 35$$

$$\therefore n \leq \frac{2 \times 35}{3}$$

$$n \leq 23$$

$$\Rightarrow n = 23 \quad \text{Ans.}$$

$K_n \rightarrow$ Complete graph with n vertices

$$\text{No. of edges in } K_n = \frac{n(n-1)}{2}$$

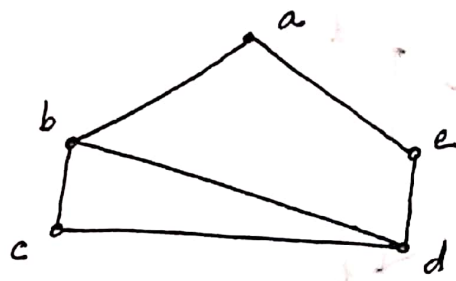
$\rightarrow$ Adjacency Matrix ($A(G)$)

$$a_{ij} = \begin{cases} 1, & \text{if } v_i \text{ and } v_j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases}$$

1. Square matrix

2. For simple graph G , no. of 1's in row =
no. of 1's in column = degree (v_i)

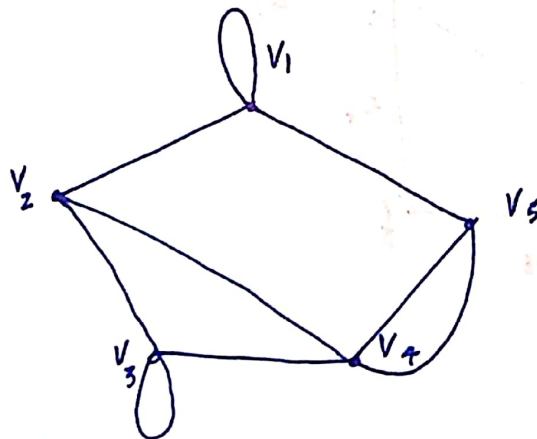
3. For simple graph G , entries on principle diagonal are zero

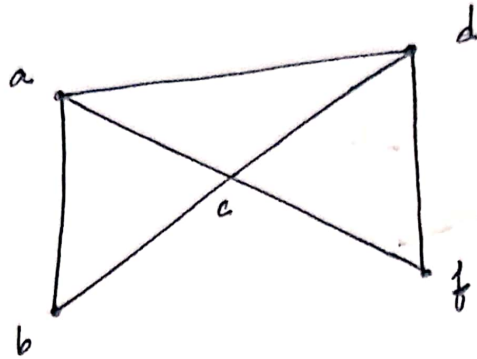


$A(G) =$

$$\begin{matrix}
 & \begin{matrix} a & b & c & d & e \end{matrix} \\
 \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}
 \end{matrix}$$

$$\begin{matrix}
 & \begin{matrix} V_1 & V_2 & V_3 & V_4 & V_5 \end{matrix} \\
 \begin{matrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 2 \\ 1 & 0 & 0 & 2 & 0 \end{bmatrix}
 \end{matrix}$$

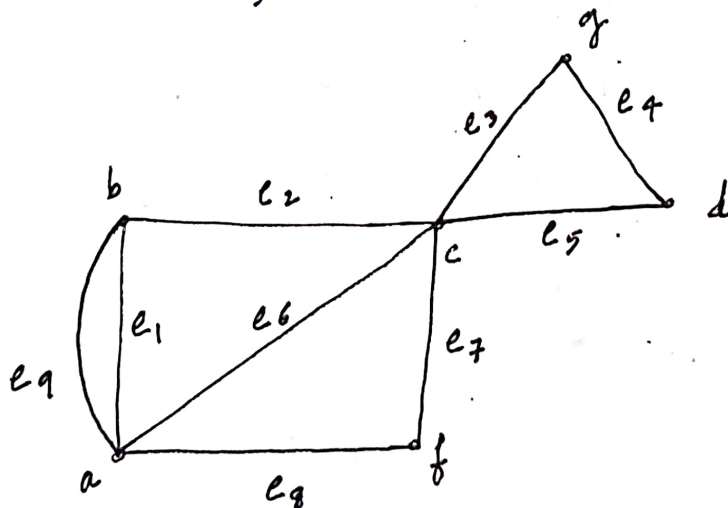




$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Incidence Matrix $X(G)$

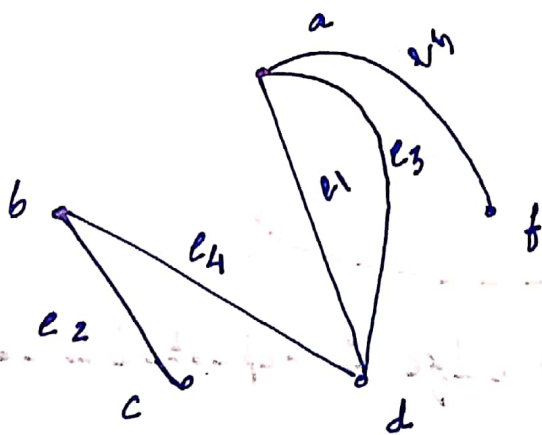
$$x_{ij} = \begin{cases} 1, & \text{if } j^{\text{th}} \text{ edge } (e_j) \text{ is incident on } i^{\text{th}} \text{ vertex } (v_i) \\ 0, & \text{otherwise} \end{cases}$$



$X(G) =$

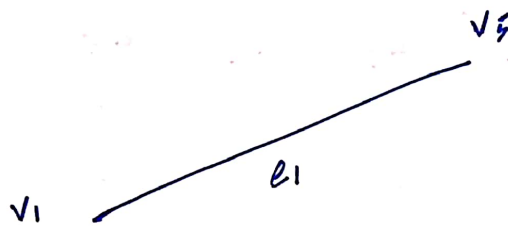
	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
a	1	0	0	0	0	1	0	1
b	1	1	0	0	0	0	0	0
c	0	1	1	0	1	1	1	0
d	0	0	0	1	1	0	0	0
e	0	0	0	0	0	0	1	1
f	0	0	1	1	0	0	0	0

	e_1	e_2	e_3	e_4	e_5
a	1	0	1	0	1
b	0	1	0	1	0
c	0	1	0	0	0
d	1	0	1	1	0
e	0	0	0	0	1



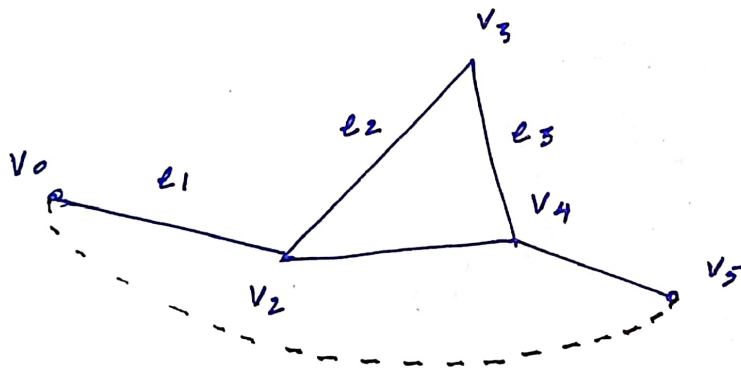
Walk - A walk in a graph is defined as finite alternating sequence of vertices and edges beginning and ending with vertices in which each edge is incident with two points only one immediately preceding and one following it.

$V_0 e_1 V_2 e_3 V_4 e_5 V_3 e_2 V_5 \dots V_5$
 Initial vertex $\nearrow$ $\nwarrow$ Terminal vertex



If $V_0 = V_n \rightarrow$ closed walk

If $V_0 \neq V_n \rightarrow$ open walk



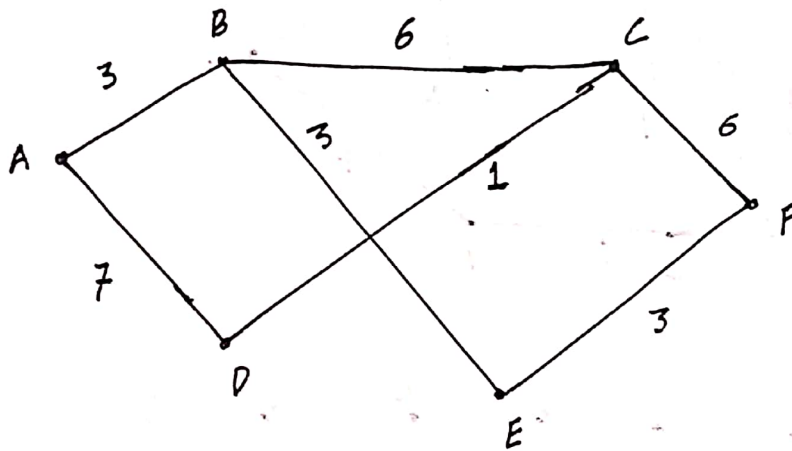
Open walk in which no edge is repeated is called as a **trail**.

$\rightarrow$ **Path**: Open walk in which no vertex appears more than once. Repetition of vertex not allowed.

$\rightarrow$ **Circuit**: A closed path is called as a circuit.

▣ Dijkstra's Algorithm:
To find the shortest path between two vertices.

Determine shortest path from A to F given this graph.



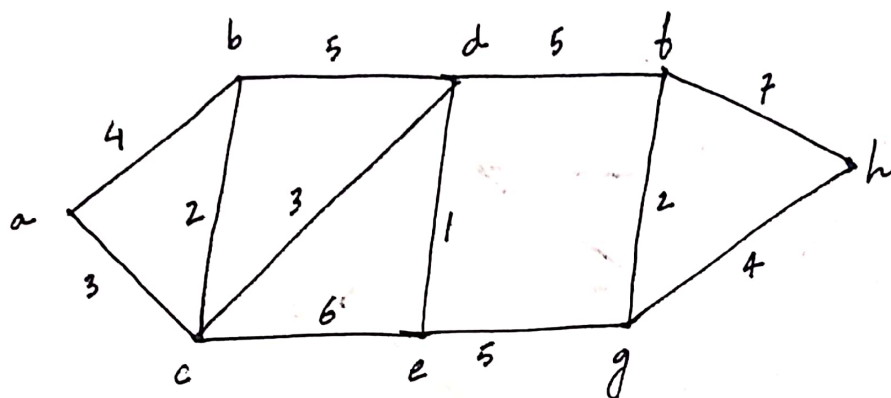
A	B	C	D	E	F
	3(ab)	∞	7(ad)	∞	∞
		8(abc)	7(ad)	abe(6)	∞
		9(abc)	7(ad)		9(abef)
		8(adc)	7(ad)		9(abef)
		8(ade)			9(abef)
					9(abef)

Shortest path : abef

Weight of path : 9

Ans.

Q. Find the shortest path from A to H using Dijkstra's Algorithm.



a	b	c	d	e	f	g	h
	4(ab)	3(ac)	∞	∞	∞	∞	∞
4(ab)			6(acd)	9(ace)	∞	∞	∞
		6(acd)		9(ace)	∞	∞	∞
			7(acde)	11(acdf)	∞	∞	
				11(acdf)		12(acdeg)	∞
					12(acdfg)		18(acdfgh)
							16(acdegh)

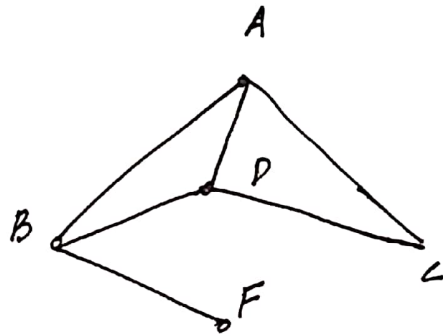
Shortest path : a c d e g h

Weight : 16 Ans.

Connected Graph

A graph G is said to be connected if every pair of points in G is joined by a path.

Component: Maximal connected subgroup of G is called a 'component'.



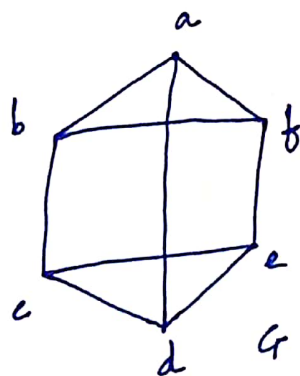
Isomorphism

Two graphs G and G' are isomorphic if there is a function 'f'.

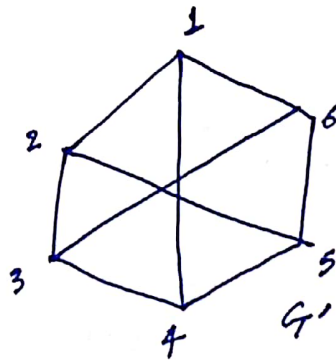
$$f: V(G) \rightarrow V(G') \text{ such that}$$

- (i) f is one-one
- (ii) f is onto
- (iii) For each pair of vertices u and v of G , $\{u, v\}$ is connected by an edge in G then $\{f(u), f(v)\} \in E(G') \Rightarrow$ adjacency is preserved.

Ex :



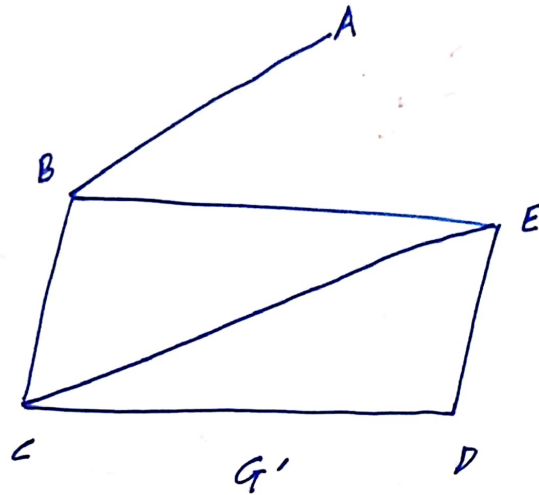
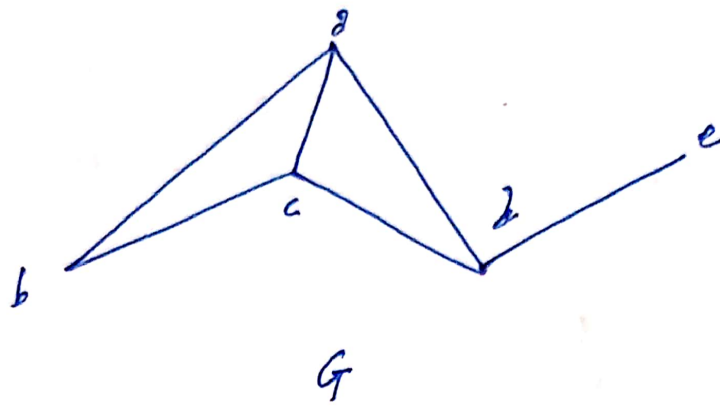
$$\begin{aligned} \therefore f(a) &= 1 & f(d) &= 4 \\ f(b) &= 2 & f(e) &= 5 \\ f(c) &= 3 \end{aligned}$$



But, as 'e' is not connected to 'b' but

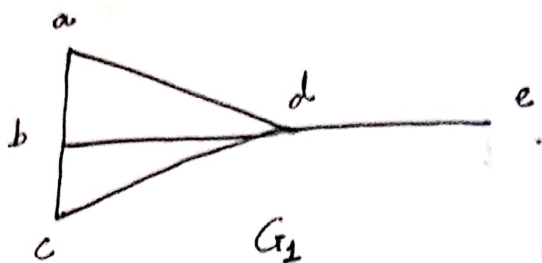
5(e) is connected to 2(b) in G'

Hence G and G' are not isomorphic.

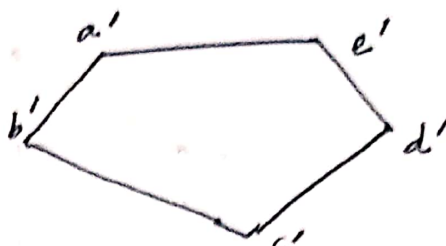


$$\therefore \begin{aligned} f(a) &= A \\ f(b) &= D \\ f(c) &= E \\ f(d) &= C \\ f(e) &= B \end{aligned}$$

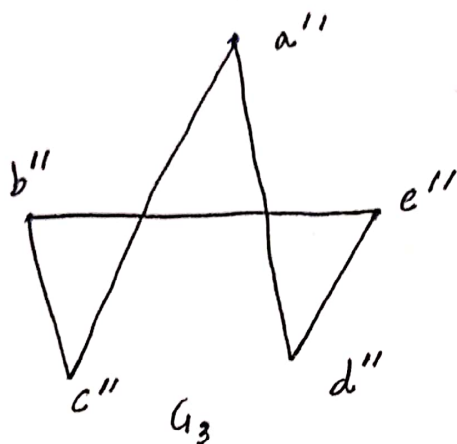
As, the mapping is successful
 G and G' are isomorphic.



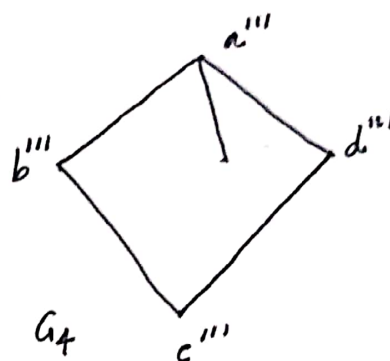
G_1



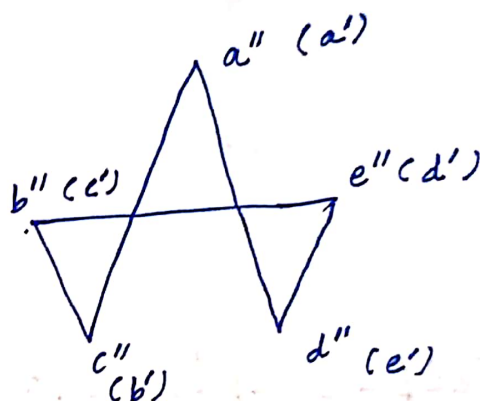
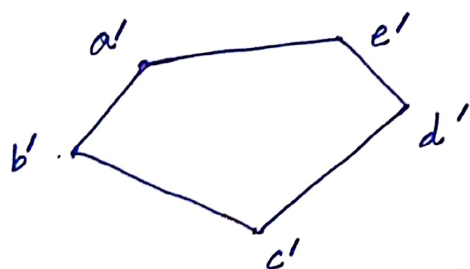
G_2



G_3



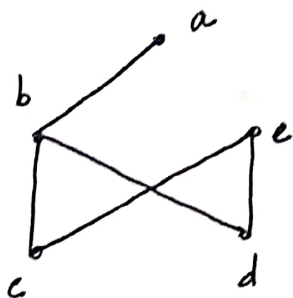
G_4



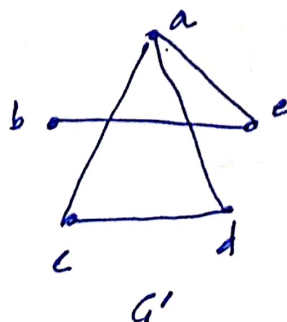
$\therefore$
 $f(a') = a''$
 $f(b') = c''$
 $f(c') = b''$
 $f(d') = e''$
 $f(e') = d''$

$\therefore G_2$ and G_3 are iso-morphic graphs.

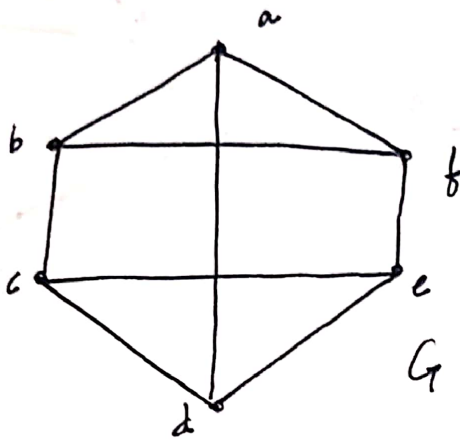
ii Complement of Graph ' G ' $\rightarrow$ Complement of G from complete graph of G .



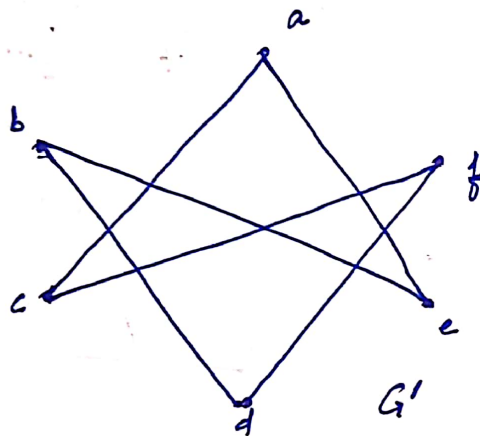
$\rightarrow$



G'



complement



Self-complementary graph

$$\#(V(G)) = \#(V(G'))$$

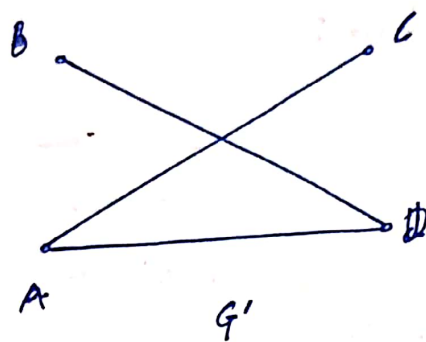
$$\#(E(G)) = \#(E(G'))$$

G and G' is an isomorphic

Statement: The graph (p, q) is self complementary if p or $(p-1)$ is divisible by 4.



4 vertices,
self complementary ✓



$$\begin{aligned} f(a) &= B \\ f(b) &= C \\ f(c) &= D \\ f(d) &= A \end{aligned}$$

$$\# \text{ Edges in self complementary graphs} = \frac{p(p-1)}{2}$$

Eulerian Circuit

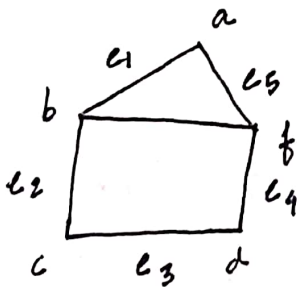
Circuit that contain all the edges of graph.

Eulerian Graph:

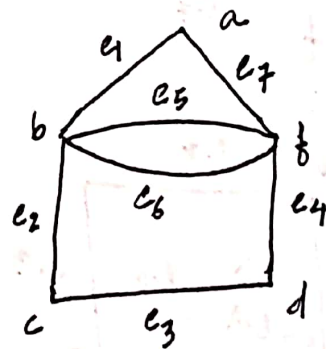
A graph containing eulerian circuit is called Eulerian graph.

(edges $\rightarrow$ no repeat, vertex may repeat)

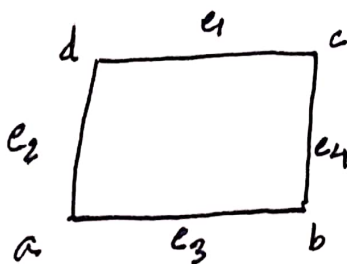
Ex :



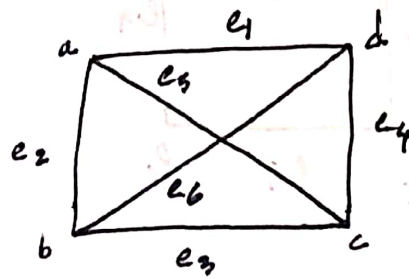
Not Eulerian



Eulerian



Eulerian



Not Eulerian

□ Fleury's Algorithm:

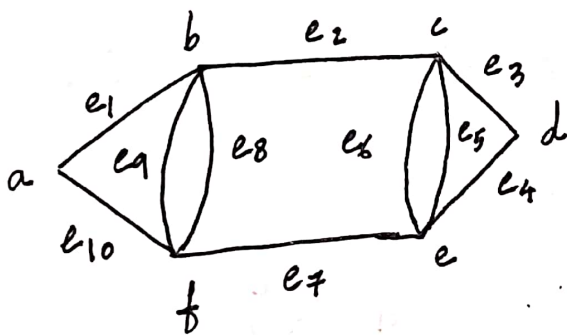
Start at any vertex and proceeding along any edges one-by-one-subject to-

(a) Erase all edges they are used and the vertex becomes isolated by doing so.

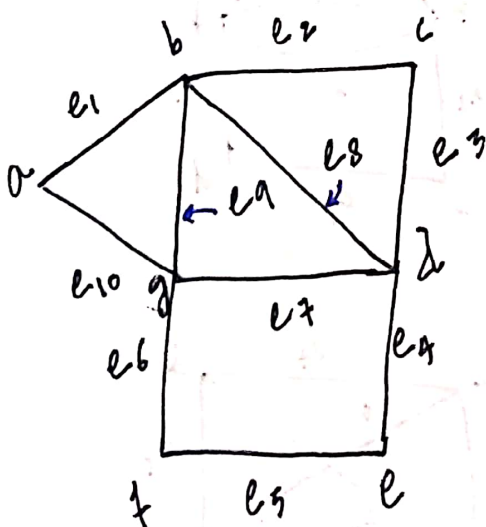
(b) Do not use an edge if its removal disconnects the graph at that point, unless there is no alternative.

(c) No edge is remaining

Q. Find Eulerian circuit using Fleury's algorithm



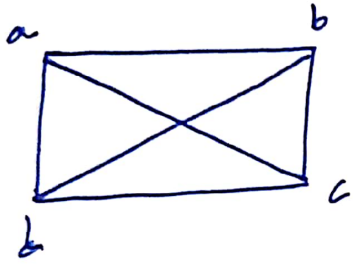
$\pi: a e_1 b e_2 c e_3 d e_4 e e_5 c e_6 e_7 f e_8 b e_9 f e_{10} a$



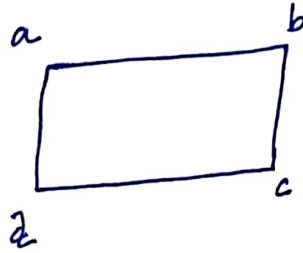
$\pi: a e_1 b e_8 d e_7 g e_9 b e_2 c e_5 d e_4 e e_5 f e_6 g e_{10} a$

- Hamiltonian Circuit: A circuit that contains all vertices of graphs called Hamiltonian circuit
- Hamiltonian Graph: Graph that includes Hamiltonian circuit

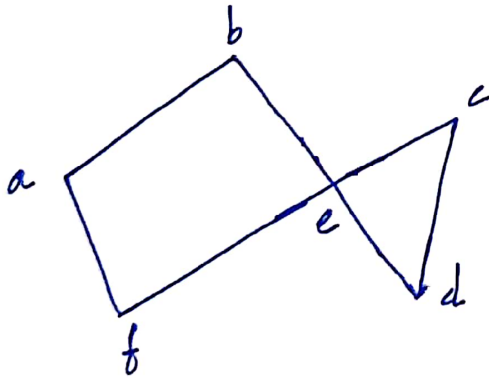
All vertices $\begin{cases} \rightarrow \text{not repeated} \\ \rightarrow \text{all edges may not be included} \end{cases}$



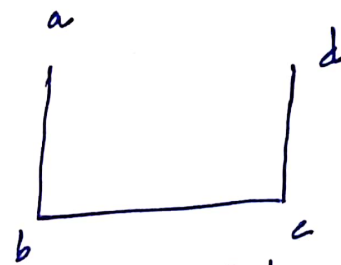
Hamiltonian graph



Both Hamiltonian and Eulerian.



Eulerian but Not Hamiltonian



Neither Eulerian nor Hamiltonian

Handwritten signature